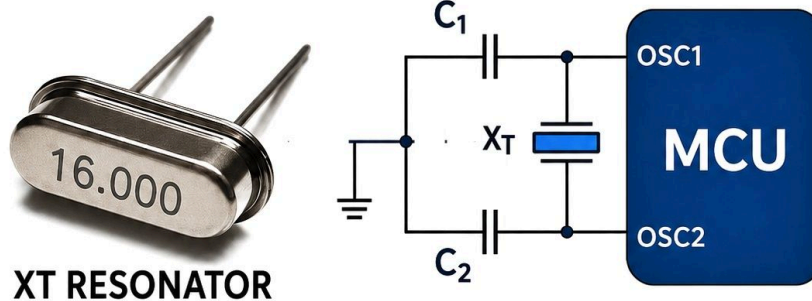




CRYSTAL OSCILLATOR

An electronic circuit using a piezoelectric crystal for highly stable frequency signals.



XT RESONATOR



PIEZOELECTRIC
RESONANCE



HIGH FREQUENCY
STABILITY



SERIES &
PARALLEL
MODES

LOAD CAPACITANCE (C_L)

$$C_L = \frac{C_1 \times C_2}{C_1 + C_2} + C_s$$

Where: C_1 , C_2 = Loading Capacitors; C_s = Stray PCB Capacitance;
 C_L = Target Load for Stable Operation.

APPLICATIONS



MICROCONTROLLER
CLOCKING



DIGITAL
COMMUNICATIONS



AUTOMOTIVE
ELECTRONICS



Electronics Education

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A crystal oscillator is a circuit that generates a very stable clock signal using the mechanical vibration of a quartz crystal. When voltage is applied to the crystal, it vibrates at a precise natural frequency because of the piezoelectric effect. This vibration creates a highly accurate electrical signal used for timing and synchronization in electronic systems.

In the diagram, the crystal resonator (XT) is connected to the microcontroller oscillator pins OSC1 and OSC2. Capacitors C_1 and C_2 help set the correct load capacitance required for stable oscillation. The MCU uses this signal as its system clock to control processing speed and timing operations.

Crystal oscillators are preferred because they provide:

- Excellent frequency stability

- Low drift with temperature changes
- Accurate timing over long periods
- Reliable operation in digital circuits

The load capacitance equation helps designers match the crystal specifications with the external capacitors for proper startup and stable performance.

Common applications include:

- Microcontroller clock generation

- Communication systems
- Automotive electronics
- Computers and embedded systems
- Wireless and RF circuits

Without a stable crystal oscillator, devices such as computers, phones, GPS systems, and communication hardware would lose timing accuracy and synchronization.

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1 v **Gilla** Svara**Esteban Viveros Reyes**

if you want to know how to calc C1 & C2:

<https://support.microchip.com/s/article/Calculating-crystal-load-capacitor>1 v **Gilla** Svara**Patoh Patoka**

Love this ❤️

1 v **Gilla** Svara**Rachid Mahamane**

Yeah 🤓

1 v **Gilla** Svara

Skriv en kommentar ...

